Business Intelligence Systems

INFO 4555 / 5556

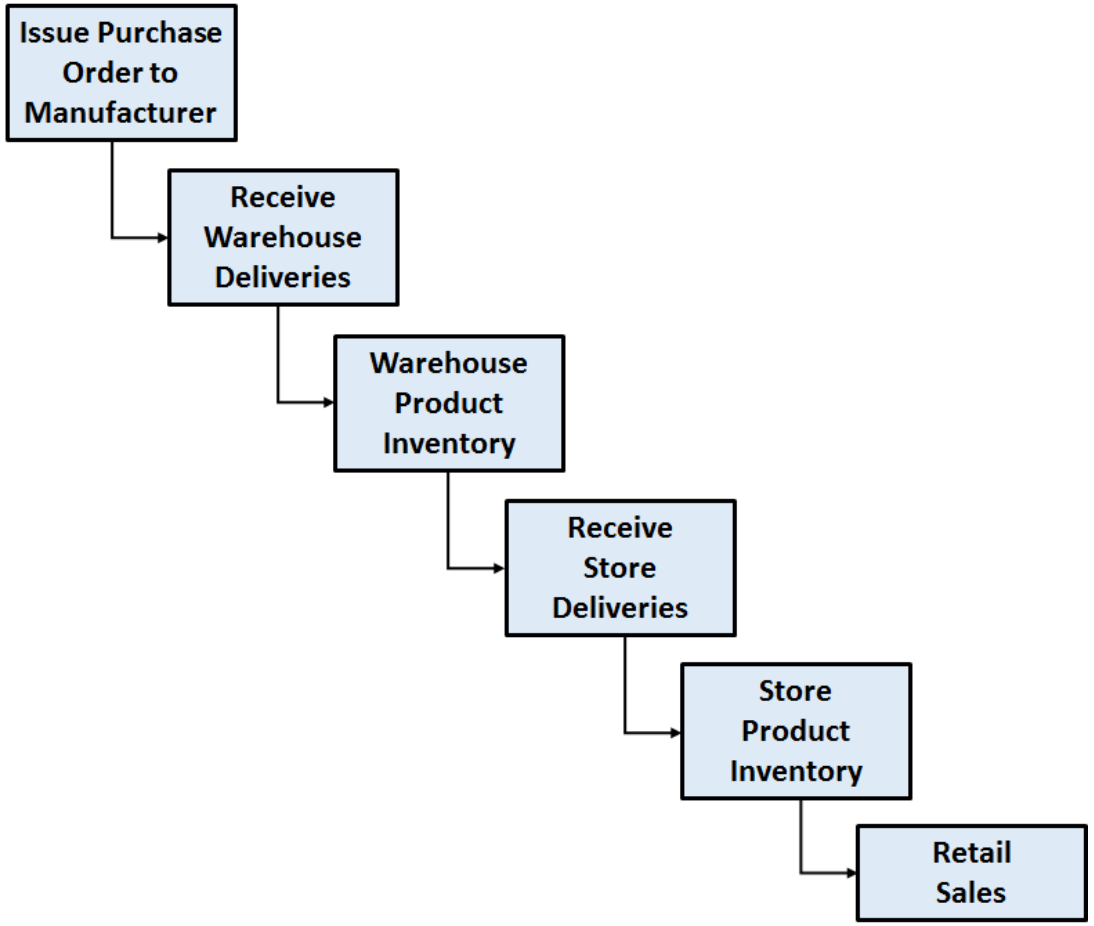
Fall 2025

Inventory



Cornell University

Kimball Value Chain Example:



The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 4-1

- Value Chain – identifies the natural logical flow of an organization's primary activities.
- Operational source systems typically produce transactions or snapshots at each step of the value chain.
- Each process in a value chain generally translates to one or more fact tables

When poll is active, respond at **pollev.com/info4555**

True or False: Operational source systems typically produce transactions or snapshots at each step of the value chain.

True

False

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Three Types of Fact Tables

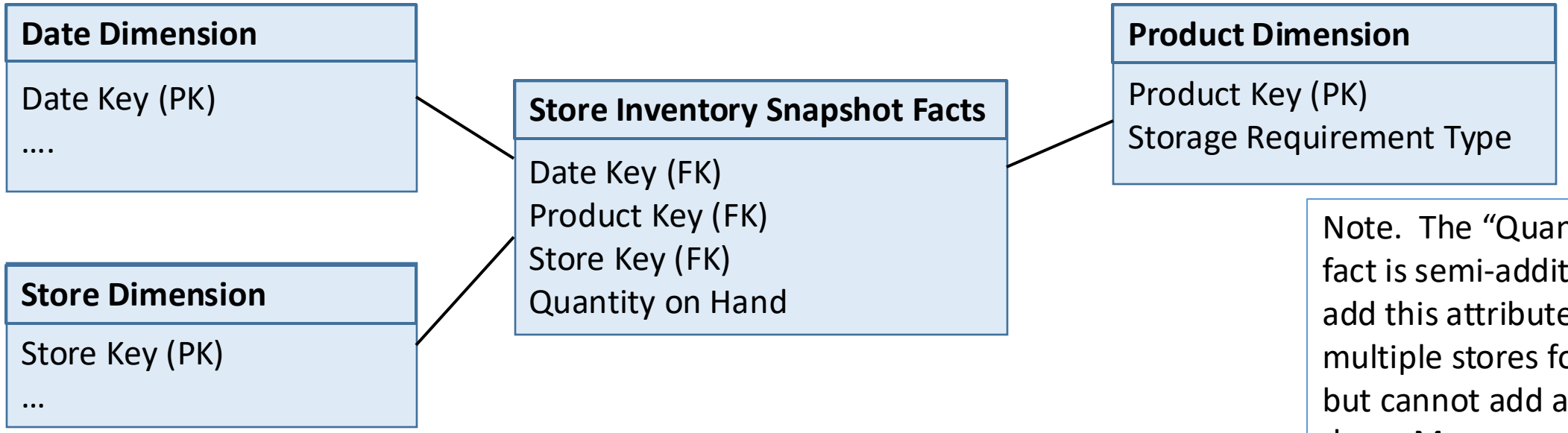
1. Transaction Fact Tables
   - Represent an event that occurred at an instantaneous point in time
   - Grain is generally at the individual transaction or transaction line level
   - Only contains records for transaction that have occurred
2. Periodic Snapshot Fact Tables
   - Represent the cumulative performance of the organization at regular, predictable intervals
   - The “snapshot” is basically a picture of the process at a point in time.
   - Common snapshot frequencies: day, week, month

Three Types of Fact Tables (continued)

3. Accumulating Snapshot Fact Tables
   - represent processes that have a definite beginning and definite end
   - also represent a common set of intermediate process steps
   - used to measure performance of workflow, or pipeline analysis
   - contain multiple date dimension foreign keys to represent milestones in a process
   - may contain “completed”, or “success” attributes (1 or 0)
   - Generally used to measure lag times between milestones in a process.
     - Where are the bottlenecks?

Kimball Inventory Periodic Snapshot example

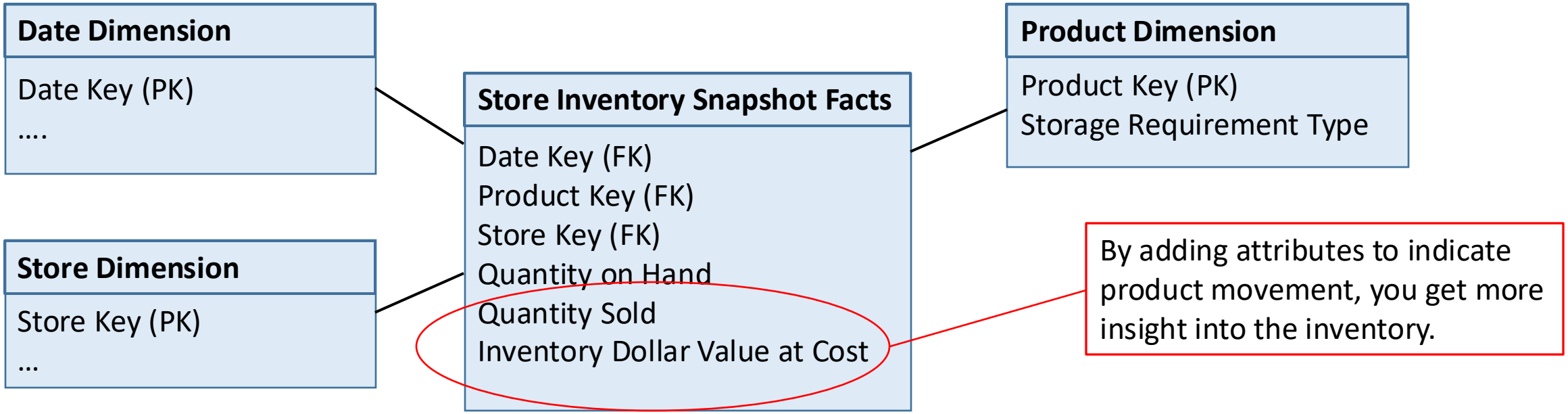
Grain: Daily inventory for each product in each store



The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 4-2

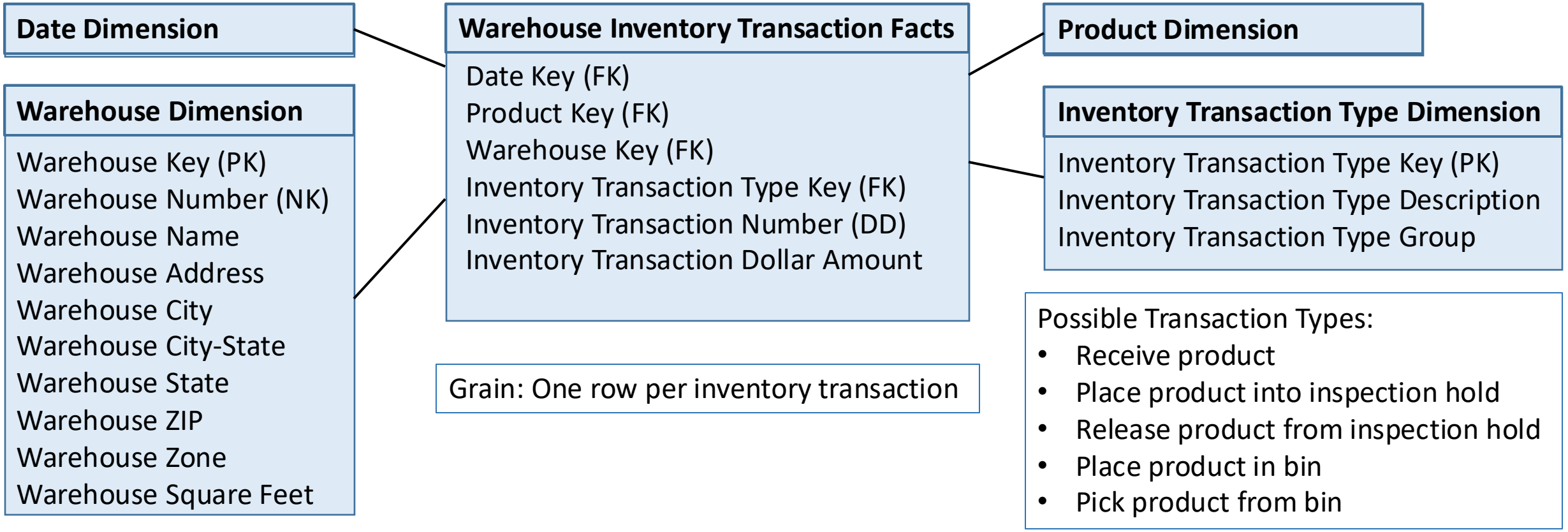
Note. The "Quantity on Hand" fact is semi-additive. You may add this attribute across multiple stores for a single day, but cannot add across multiple days. May average this attribute over time.

Kimball Enhances Inventory Periodic Snapshot



The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 4-3

Kimball Inventory Transaction example



The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 4-4

When poll is active, respond at **pollev.com/info4555**

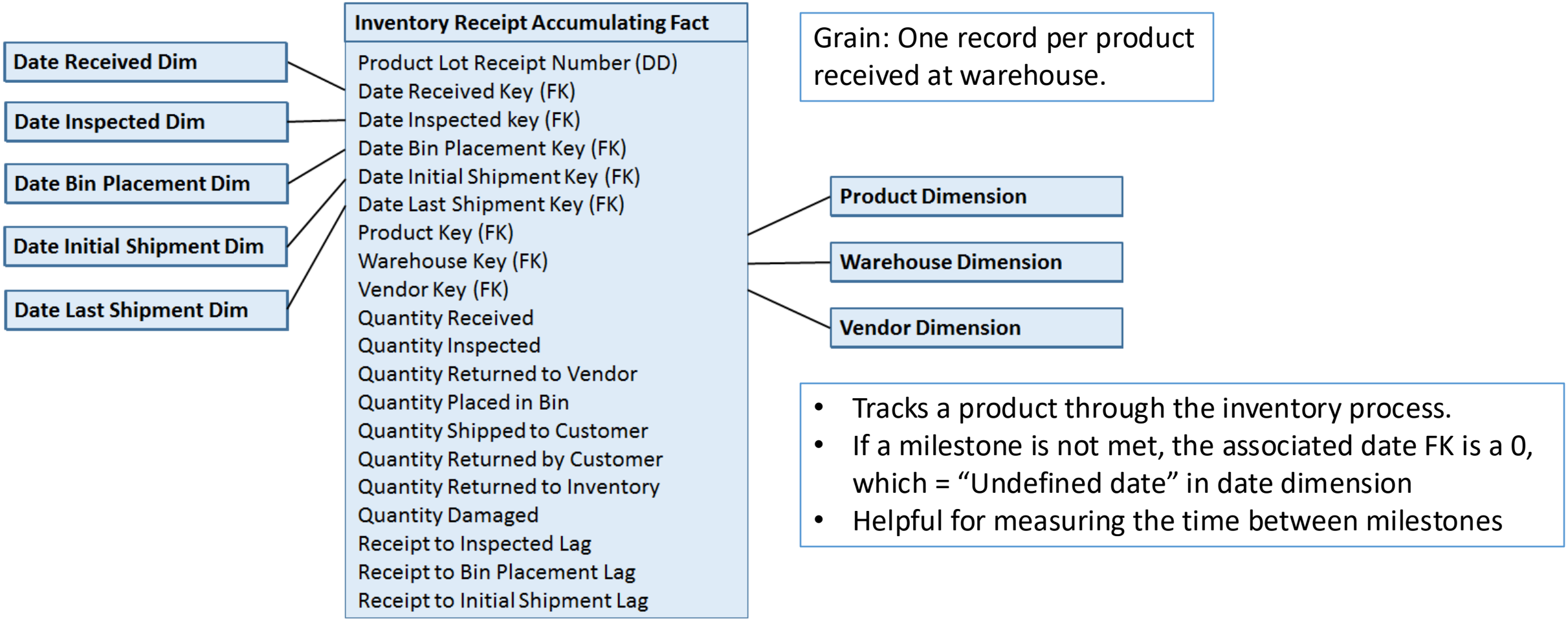
Text **INFO4555** to **22333** once to join

The following statement is true about which type of fact table? Often the only place to easily retrieve a regular, predictable view of longitudinal performance trends.

A: Type I
B: Periodic snapshot
C: Aggregate
D: Accumulating snapshot

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Kimball Inventory Accumulating Snapshot



Grain: One record per product received at warehouse.

- Tracks a product through the inventory process.
- If a milestone is not met, the associated date FK is a 0, which = "Undefined date" in date dimension
- Helpful for measuring the time between milestones

The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 4-5

Kimball Inventory Accumulating Snapshot (continued)

Fact row inserted when lot received:

Lot Receipt Number	Date Received Key	Date Inspected Key	Date Bin Placement Key	Product Key	Quantity Received	Receipt to Inspected Lag	Receipt to Bin Placement Lag
101	20130101	0	0	1	100		

Fact row updated when lot inspected:

Lot Receipt Number	Date Received Key	Date Inspected Key	Date Bin Placement Key	Product Key	Quantity Received	Receipt to Inspected Lag	Receipt to Bin Placement Lag
101	20130101	20130103	0	1	100	2	

Fact row updated when lot placed in bin:

Lot Receipt Number	Date Received Key	Date Inspected Key	Date Bin Placement Key	Product Key	Quantity Received	Receipt to Inspected Lag	Receipt to Bin Placement Lag
101	20130101	20130103	20130104	1	100	2	3

The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 4-6

When poll is active, respond at **pollev.com/info4555**
Text **INFO4555** to **22333** once to join

The three types of fact tables are:

A: Transaction, periodic snapshot, accumulating snapshot

B: Historical, periodic snapshot, Transactional

C: Metric, accumulating snapshot, aggregating

D: Type I, Type II, Type III

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Database TABLES versus VIEWS

- TABLES are physical objects which contain records
- VIEWS are virtual objects which DO NOT contain data, but reference data in TABLES
  - A searchable object that is defined by a query
  - May contain data from multiple tables, through a table join in the defining query
  - May contain a subset of data through a filter in the defining query
  - Attribute names may differ from the underlying tables, depending on the VIEW definition. **This is key for dimension role playing.**



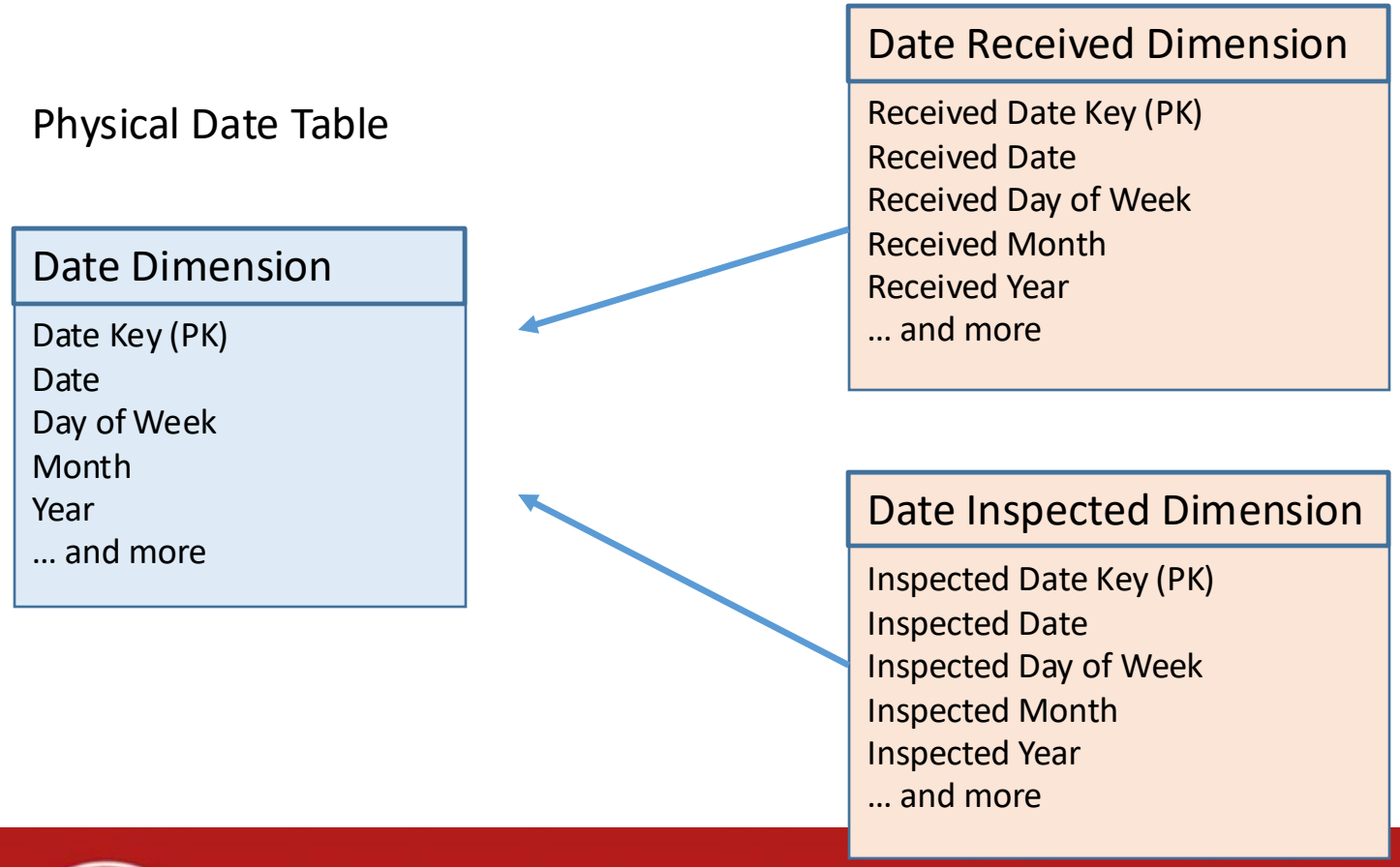
Role Playing Dimensions

- The use of database VIEWS to represent a single dimension table for differing logical purposes
- Commonly used for Date Dimensions to allow a FACT table to reference multiple dates on a single record.

Role Playing Dimensions Example

Views of Date Table with varying View and Attribute names

Physical Date Table



- Notice the differing names of views and their attributes compared to the underlying Date Dimension Table.
- These Date Views are Role Playing.

Role Playing Dimensions (Sample SQL)

```
CREATE VIEW Date_Received_Dimension
(Received_Date_Key, Received_Date, Received_Day_of_Week,
Received_Month, Received_Year)
AS
SELECT Date_Key, Date, Day_of_Week, Month, Year
FROM Date_Dimension;
```



When poll is active, respond at **pollev.com/info4555**

Text **INFO4555** to **22333** once to join

Which of the following statements is TRUE regarding Dimension Role playing?

A: Not recommended in STAR schemas

B: Allows a dimension to be used more than once in a STAR schema without confusing the context

C: Results in doubling the storage of the master dimension

D: Should have a unique Primary Key for each “role play” dimension

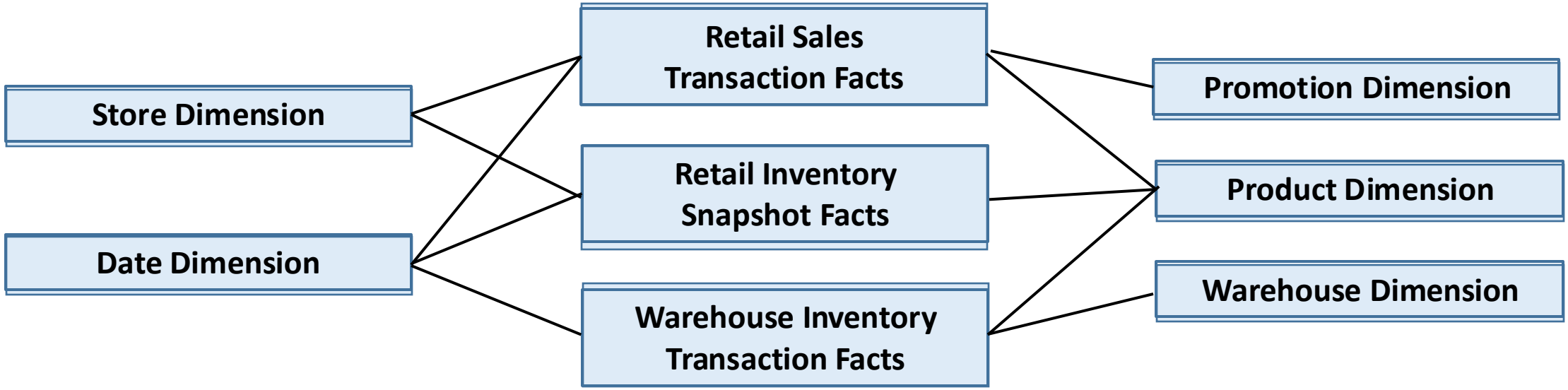
Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Kimball Fact Table Type Comparison

	Transaction	Periodic Snapshot	Accumulating Snapshot
Periodicity	Discrete Transaction point in time	Recurring snapshots at regular, predictable intervals	Indeterminate time span for evolving pipeline/workflow
Grain	1 row per transaction or transaction line	1 row per snapshot period plus other dimensions	1 row per pipeline occurrence
Date Dimension(s)	Transaction date	Snapshot date	Multiple dates for pipeline's key milestones
Facts	Transaction performance	Cumulative performance for time interval	Performance for pipeline occurrence
Fact table sparsity	Sparse or dense, depending on activity	Predictably dense	Sparse or dense, depending on pipeline occurrence
Fact table updates	No updates, unless error correction	No updates, unless error correction	Updated whenever pipeline activity occurs

The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 4-7

Sharing Dimensions among Business Processes



The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 4-8

Enterprise Data Warehouse Bus Matrix

A tool to document shared dimensions and the business processes (FACT tables) which leverage them.

Sample Bus Matrix

	Common dimensions					
Metrics / Processes	Date	Person *	Org*	Data Mart	OBIEE Groups	Table
Load Success	X			X		
Data Mart Records	X			X		X
Load Times	X			X		
OBIEE Usage	X	X	X		X	
OBIEE Response time	X	X	X		X	
	*Org may simply be an attribute on the Person dimension					



When poll is active, respond at **pollev.com/info4555**
Text **INFO4555** to **22333** once to join

The Enterprise Data Warehouse Bus Architecture:

A: Is a cost effective way to travel to class

B: Allows users to plug their smart phone directly into the data warehouse

C: Allow dimensional models to integrate across subject areas

D: Is often referred to as the Big Data approach.

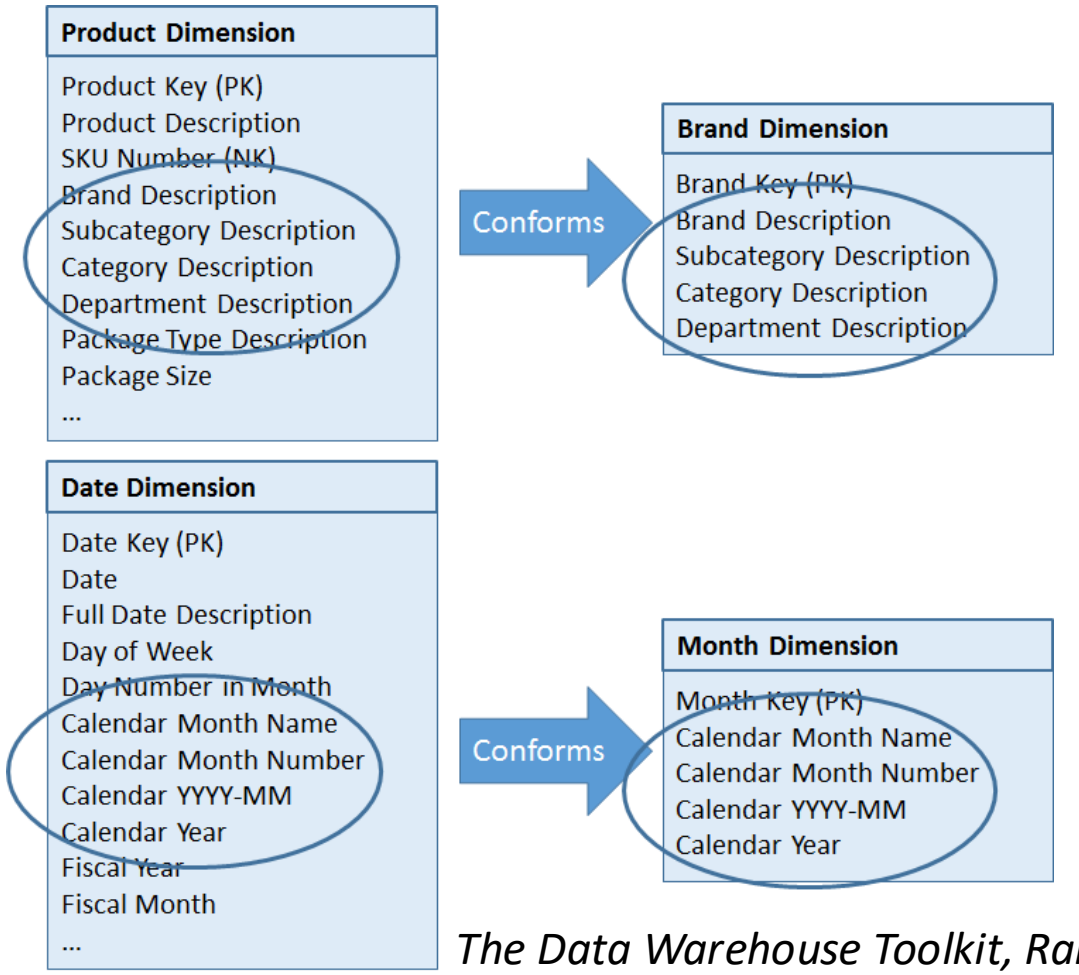
Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Conformed Dimensions

- Primary key is the same for the dimension, regardless of where it is located
- Attributes match across like dimensions
- Same values and number of records
- Key to cross functional area analytics (Enterprise Data Warehouse)
- May use “Shrunken Conformed Dimensions” if only a subset is needed, but the attributes used must match the master dimension.



Kimball Sample Shrunken Conformed Dimensions:



- Depending on the grain of the fact table, a subset of the Dimension attributes or records may be appropriate
- The attribute names, values and keys must match the master Dimension in order to be conformed

The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 4-13

What is Data Governance?

Data Governance (DG) is the overall management of the availability, usability, integrity and security of data used in an enterprise.

searchdatamanagement.techtarget.com/definition/data-governance

Data Governance (DG) is the practice of organizing and implementing policies, procedures and standards for the effective use of an organization's structured/unstructured information assets.

www.sdn.sap.com

Data Governance

- A critical role of Data Governance is the definition of Conformed Attributes at the Enterprise level
- Conformed Dimensions and the ability to create an Enterprise Data Warehouse relies on Data Governance and the ability to agree on common definitions for core shared attributes
- Agreement on attribute definitions can be very difficult
- Attribute definitions traditionally done at departmental level, but in order to avoid departmental silos, it must be done at Enterprise level

Data Governance (continued)

- Data Governance body should include key stake holders (owners of operational systems, consumers of data, IT, etc)
- Lack of strong Data Governance results in:
  - Inconsistent labeling of attributes and inconsistent reporting results between department systems
  - Inability to produce consistence enterprise reporting results

When poll is active, respond at **pollev.com/info4555**

Text **INFO4555** to **22333** once to join

All of the following are TRUE about Data Governance except:

A: Should include Stake Holders from each affected business unit

B: Primary purpose is to define attributes at the Enterprise level

C: Generally well adopted by departments

D: Seeks to avoid the misrepresentation of data between departments

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/info4555**

Text **INFO4555** to **22333** once to join

How are you feeling about the material from Chapter 4?

A: Good – Bring on Chapter 5

B: OK – It kind of makes sense.

C: Bad – This is not making sense.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**